

Diagram illustrating the quotient ring $E_2 = K[x, y]/(y^2)$. The grid shows the basis elements $x^p y^q$ for $q \in \{0, 1\}$ and $p \in \{0, 1, 2, 3, 4, \dots, 2n-2\}$. Red arrows indicate the relation $y^2 = 0$, showing that $x^p y^2 = 0$ for all p . Green wavy lines separate the columns at $p=4$ and $p=2n-2$. The final entry is $x^{2n-1} = 0$.